

Patrick Tinsley

1125 Maxwell Lane • Hoboken, NJ 07030 • (650) 862-8977 • patrick.g.tinsley@gmail.com
LinkedIn • Google Scholar • GitHub

EDUCATION

University of Notre Dame <i>PhD in Computer Science and Engineering</i> Dissertation: "Trust, AI, and Synthetic Biometrics" (Advised by Dr. Pat Flynn and Dr. Adam Czajka)	South Bend, IN August 2018 - May 2024
University of Notre Dame <i>MSc in Applied and Computational Mathematics and Statistics</i> Concentration in Predictive Analytics (Advised by Dr. Steven Buechler)	South Bend, IN August 2017 - May 2018
University of Notre Dame <i>BSc in Applied and Computational Mathematics and Statistics</i>	South Bend, IN August 2013 - May 2017

PROFESSIONAL EXPERIENCE

AngelEye Health Manager of Medical Technology	Nashville, TN May 2024 - Present
- Developed a real-time alert system to notify NICU nurses when an infant may be at risk of falling from their crib; mat-based pressure sensor minimizes power consumption and data transfer volume. - Coupled a 2D pose estimate pipeline with visualization tools to aid human assessors in detecting cramp-synchronized movements patterns in videos of premature babies. - Partnered with Nationwide Children's Hospital to augment video data with pressure sensor data for the 5,000 Babies Project; conducted extensive pattern analysis on pose estimate data from 800 videos in order to detect abnormal movements; handcrafted features for rigorously evaluating symmetry and cohesion in limb movement for both camera and mat data.	
University of Notre Dame Research/Teaching Assistant	South Bend, IN August 2018 - May 2024
- Introduced and quantitatively assessed the phenomenon of <i>identity leakage</i> in biometric image generation, wherein AI-synthesized images falsely match authentic images seen during training; experiments covered three major biometric modalities: face, iris, and fingerprint. - Designed and performed matching experiments with 5 state-of-the-art face encoders to evaluate latent space clustering tendencies in the data context of large-scale, racially-diverse, pose-variant face images. - Implemented a progressive web app to interactively conduct tests with and visualize results from person re-identification models for 12 in-the-wild CCTV surveillance videos; project deliverable allowed security officers to upload and identify persons of interest in long-play video streams as well as flag false positive matches for later model re-training. - Adapted in-class coding practicals, homework assignments, and placement tests to be automatically graded via Vocareum; manually graded remaining assessments on a weekly basis for 50+ students.	

Aunalytics Machine Learning Intern	South Bend, IN August 2018 - May 2024
- Benchmarked training times for 150+ regression-based models that varied in network architecture, loss function, and size of training data; developed linear model to estimate turnaround time for training B2B models given desired model and data specifications. - Organized and maintained company's internal AI model zoo of production-ready inference models.	

SKILLS & AWARDS

Programming: Python (TensorFlow, Keras, PyTorch, Jupyter, scikit-learn, SciPy), R, Matlab, C/C++, SQL.

Miscellaneous: Git, Tableau, bash, LaTeX, Google Suite, Microsoft Office.

Honors & Awards: 2018 Graduate Student Union Outstanding TA Winner (Data Science), 2016-2017 Student Union Board Director of Publicity.